

Exploring Equity in Twin Cities: A Spatial Analysis of Ridership and Demographics

Does Metro Transit serve the people who
need it most?



Research Question: Does Twin Cities transit adequately serve marginalized communities?

- Does income predict how much transit service a neighborhood gets?
- Do communities of color have better or worse access?
- Is service quality (wait times, hours) equitable, not just coverage?

Defining Accessibility:

Having a bus route nearby does not equate having good transit.



Data Characteristics and Key Terminology

01 **Metro Transit shapefile (MN Geospatial Commons)**
75,634 road segments with trip counts and headways across 7 service periods

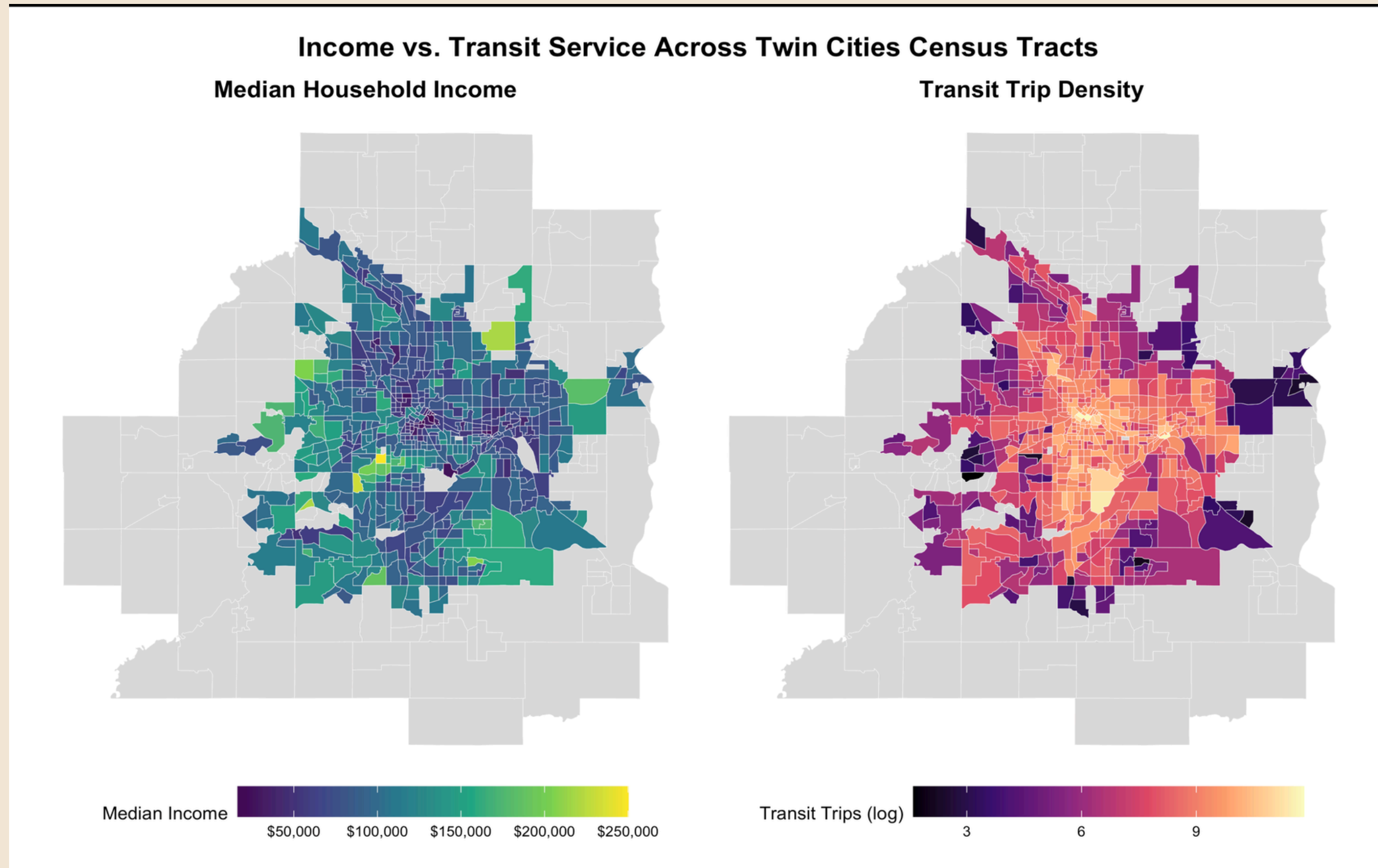
02 **ACS 2022**
Median household income and racial demographic data for all MN census tracts

Headway: minutes between consecutive buses.

TripCount: total scheduled trips per segment.

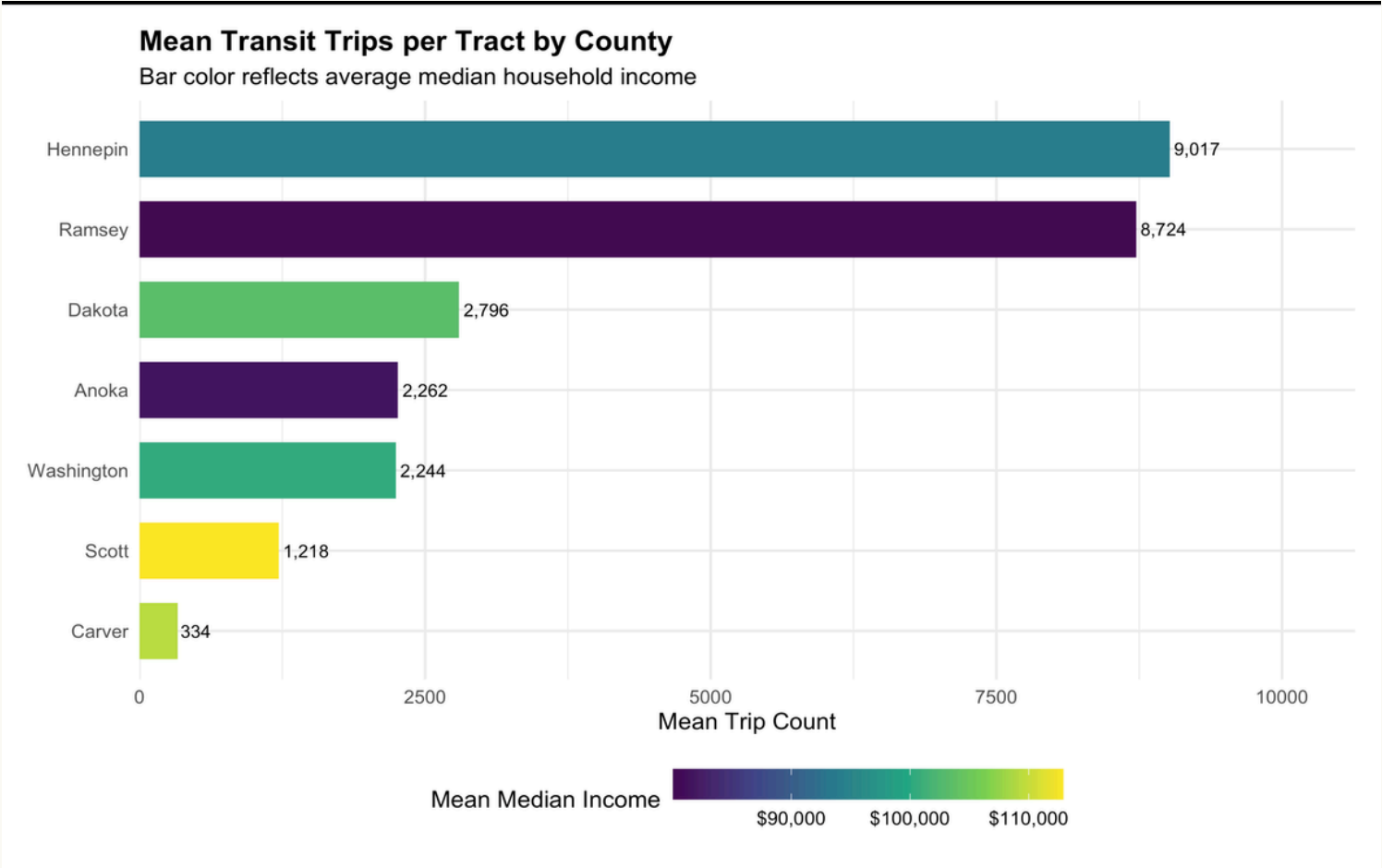
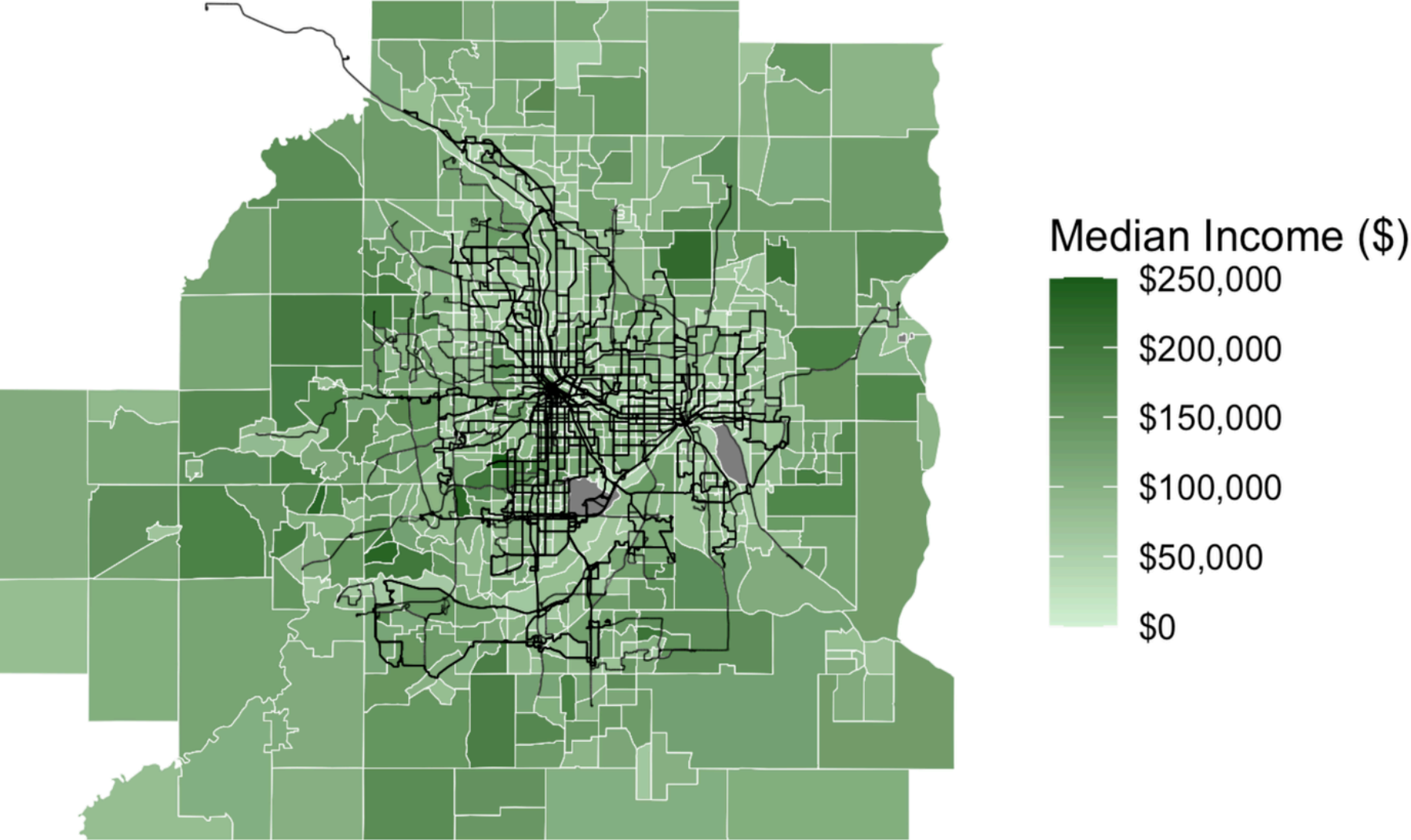


Low income areas have dense transit networks



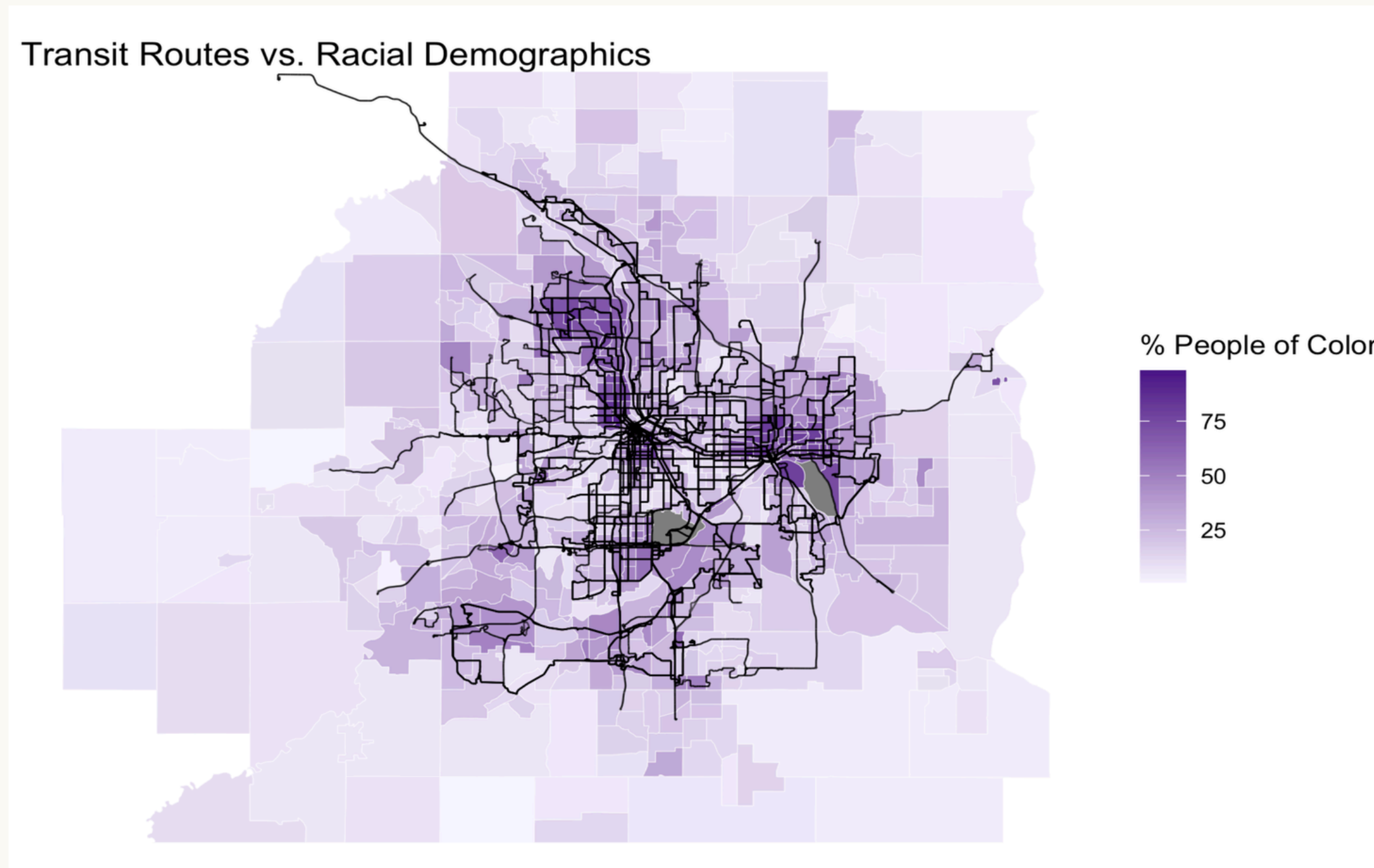
Median income is lowest where transit density is highest

Transit Routes vs. Median Household Income



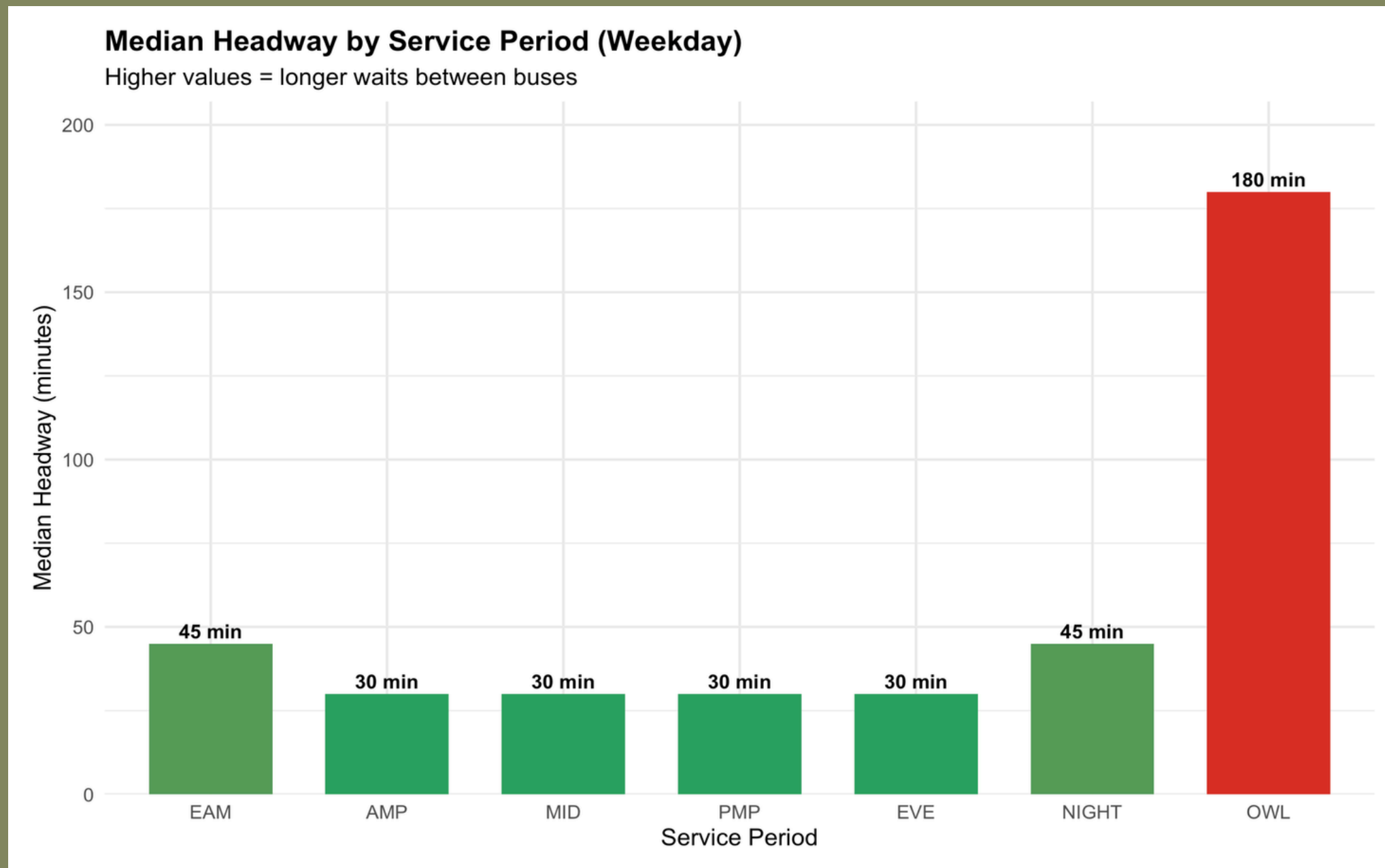
Transit service is heavily concentrated in lower-income central city tracts, while wealthier suburban areas have little to no coverage

Racially Minoritized areas also have dense transit networks



The map shows transit routes are heavily concentrated in urban core, which also has the highest density of People of Color. Suburban areas are predominantly white and have lesser transit accessibility.

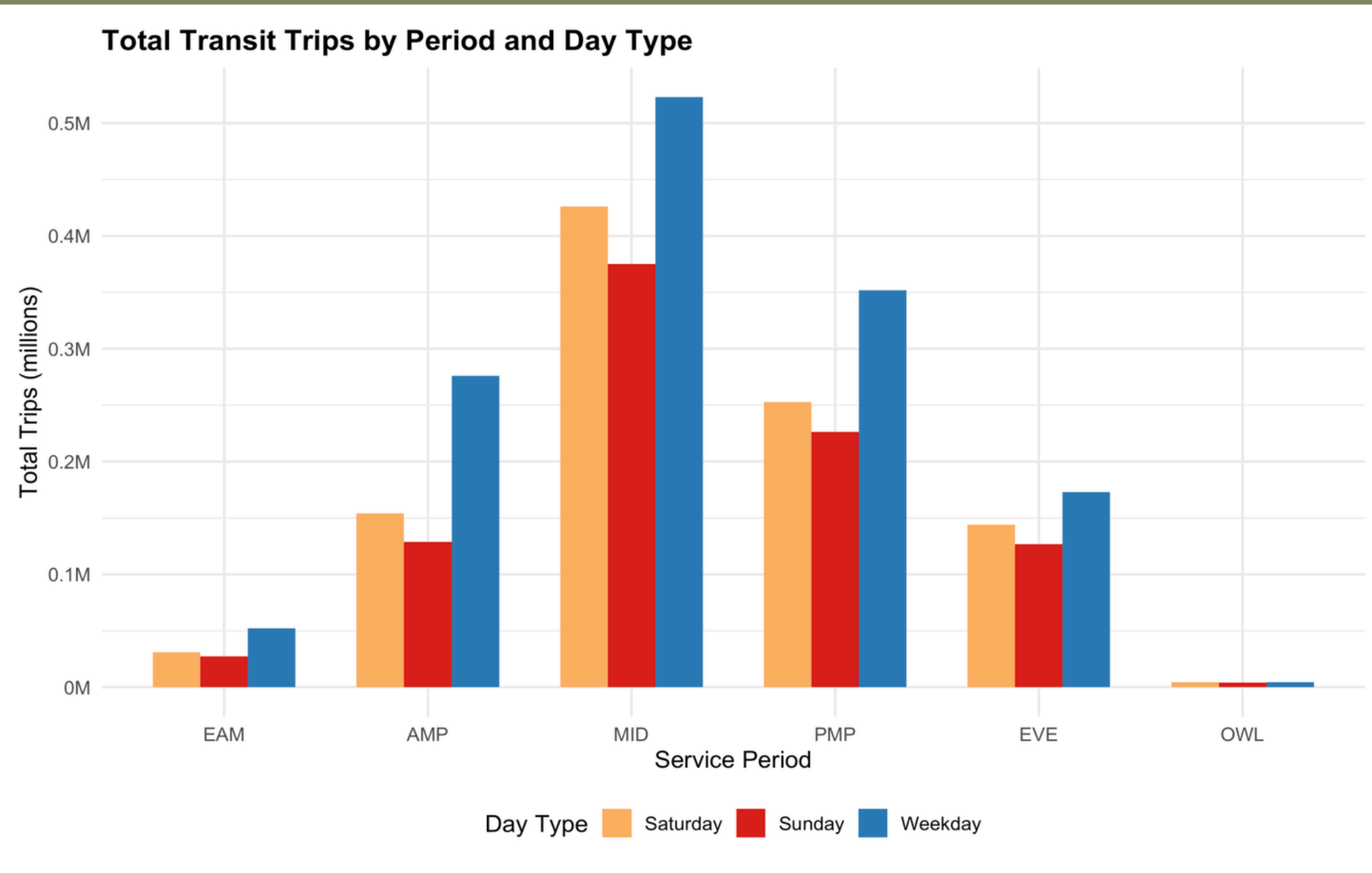
"Served" ≠ "Well-Served": The network is built for 9-to-5 commuters



Headways are longest for “owl” services.

Lower-wage workers in healthcare, hospitality, service industries, often work night shifts.

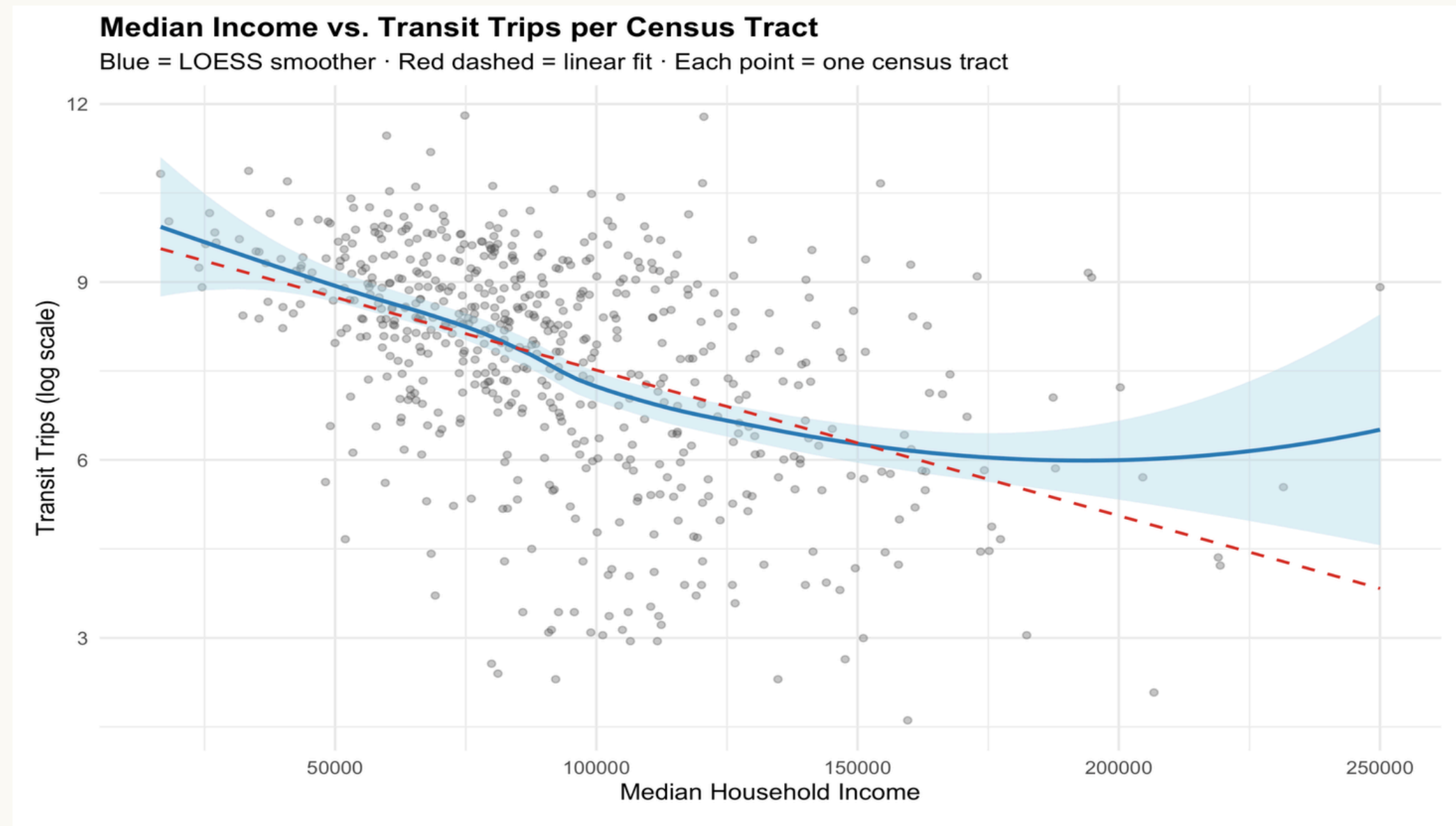
Weekend service is smoother but covers less



Weekend services have fewer routes, wider headways

Lower-income riders are more likely to work non-standard hours and depend on weekend transit.

Income Predicts Transit Access, But Imperfectly



Lower-income tracts tend to have more transit service, but the relationship is weak. Density, route history, and corridor geography are equally powerful predictors.

Conclusion and Limitations

The verdict: geographically yes, temporally no

Routes concentrate in lower-income, higher-POC neighborhoods

But service degrades sharply outside peak hours, when lower-wage workers need it most

A 90-minute mean night headway is not reliable transportation

Limitations:

Supply $\neq$ ridership; spatial join methodology introduces measurement error

22% of tracts missing income data;

Lack of multivariate analysis

Future goals:

Job accessibility metrics, longitudinal service changes, peer city comparisons

Thank you!